EECE 5644 Fall 2026 - Assignment 1

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Files: Link

Generative-AI Disclosure

I used AI tools as a support resource to help troubleshoot programming issues, clarify implementation steps, and resolve errors encountered while developing and testing my solution. I also used AI to help verify my understanding of the classification methods and improve the organization of my work. AI was also used to help me structure the writeup.

Answer 1:

Reproducibility and Dataset Generation

All numerical results in this document were generated from one fixed data set of 10,000 samples using NumPy's random number generator with seed 42. The same samples and true labels are reused in Parts A, B, and C, as required by the assignment. The exact data set is supplied separately as EECE5644_Q1_data_seed42.csv and EECE5644_Q1_data_seed42.npz, and the complete Python program is supplied as EECE5644_Q1_solution.py.

The generated class counts are

$$N = 10000, \qquad N_0 = 3457, \qquad N_1 = 6543.$$

The expected counts based on the priors are

$$E[N_0] = 10000(0.35) = 3500, \qquad E[N_1] = 10000(0.65) = 6500.$$

The small differences are due to random sampling.

Given Probability Model and Generated Dataset

The feature vector is four-dimensional:

$$\mathbf{x} \in \mathbb{R}^4.$$

The class priors are

$$P(L = 0) = 0.35, \qquad P(L = 1) = 0.65.$$

The mixture density is

$$p(\mathbf{x}) = p(\mathbf{x} \mid L = 0)P(L = 0) + p(\mathbf{x} \mid L = 1)P(L = 1).$$

The two class-conditional densities are Gaussian:

$$p(\mathbf{x} \mid L = 0) = g(\mathbf{x} \mid \mathbf{m}_0, C_0),$$

$$p(\mathbf{x} \mid L = 1) = g(\mathbf{x} \mid \mathbf{m}_1, C_1).$$

The mean vectors are

$$\mathbf{m}_0 = \begin{bmatrix} -0.5 \\ -0.5 \\ -0.5 \\ -0.5 \end{bmatrix}, \qquad \mathbf{m}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

The class-0 covariance matrix is specified as

$$C_0 = \frac{1}{4}\begin{bmatrix} 2 & -0.5 & 0.3 & 0 \\ -0.5 & 1 & -0.5 & 0 \\ 0.3 & -0.5 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

After multiplying every entry by $1/4$,

$$C_0 = \begin{bmatrix} 0.5 & -0.125 & 0.075 & 0 \\ -0.125 & 0.25 & -0.125 & 0 \\ 0.075 & -0.125 & 0.25 & 0 \\ 0 & 0 & 0 & 0.5 \end{bmatrix}.$$

The class-1 covariance matrix is

$$C_1 = \begin{bmatrix} 1 & 0.3 & -0.2 & 0 \\ 0.3 & 2 & 0.3 & 0 \\ -0.2 & 0.3 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

Preliminary Formula - Multivariate Gaussian PDF

For a d-dimensional Gaussian random vector, the generic density is

$$g(\mathbf{x} \mid \mathbf{m}, C) = \frac{1}{(2\pi)^{d/2}|C|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x} - \mathbf{m})^T C^{-1}(\mathbf{x} - \mathbf{m})\right].$$

Here $d = 4$, so

$$g(\mathbf{x} \mid \mathbf{m}, C) = \frac{1}{(2\pi)^{2}|C|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x} - \mathbf{m})^T C^{-1}(\mathbf{x} - \mathbf{m})\right].$$

Part A - ERM Classification Using the True Data PDF

A.1 - Minimum Expected-Risk Classification Rule

Let λ_{ij} denote the loss for deciding $D = i$ when the true class is $L = j$.

The conditional risk of deciding class 0 is

$$R(0 \mid \mathbf{x}) = \lambda_{00} P(L = 0 \mid \mathbf{x}) + \lambda_{01} P(L = 1 \mid \mathbf{x}).$$

The conditional risk of deciding class 1 is

$$R(1 \mid \mathbf{x}) = \lambda_{10} P(L = 0 \mid \mathbf{x}) + \lambda_{11} P(L = 1 \mid \mathbf{x}).$$

The minimum expected-risk rule selects class 1 when

$$R(1 \mid \mathbf{x}) < R(0 \mid \mathbf{x}).$$

Substituting the two risks gives

$$\lambda_{10} P(0 \mid \mathbf{x}) + \lambda_{11} P(1 \mid \mathbf{x}) < \lambda_{00} P(0 \mid \mathbf{x}) + \lambda_{01} P(1 \mid \mathbf{x}).$$

Move terms associated with the same posterior to the same side:

$$(\lambda_{10} - \lambda_{00}) P(0 \mid \mathbf{x}) < (\lambda_{01} - \lambda_{11}) P(1 \mid \mathbf{x}).$$

Using Bayes' rule,

$$P(L = j \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid L = j) P(L = j)}{p(\mathbf{x})}.$$

The common denominator $p(\mathbf{x})$ cancels, giving the likelihood-ratio test

$$\frac{p(\mathbf{x} \mid L = 1)}{p(\mathbf{x} \mid L = 0)} \gtrless_{D=0}^{D=1} \gamma,$$

where the generic ERM threshold is

$$\boxed{\gamma = \frac{(\lambda_{10} - \lambda_{00}) P(L = 0)}{(\lambda_{01} - \lambda_{11}) P(L = 1)}}.$$

A.1.1 Minimum-Probability-of-Error Threshold (0-1 Loss)

For minimum probability of error, use 0-1 loss:

$$\lambda_{00} = \lambda_{11} = 0, \qquad \lambda_{10} = \lambda_{01} = 1.$$

Substituting these values into the generic threshold gives

$$\gamma^* = \frac{(1 - 0)(0.35)}{(1 - 0)(0.65)}.$$

Therefore,

$$\boxed{\gamma^* = \frac{0.35}{0.65} = 0.5384615385}.$$

A.1.2 - Gaussian Log-Likelihood-Ratio Formula

For Gaussian class-conditionals,

$$\ell(\mathbf{x}) = \ln g(\mathbf{x} \mid \mathbf{m}_1, C_1) - \ln g(\mathbf{x} \mid \mathbf{m}_0, C_0).$$

After cancellation of the common (2π) term,

$$\boxed{\ell(\mathbf{x}) = -\frac{1}{2}\ln\frac{|C_1|}{|C_0|} - \frac{1}{2}[(\mathbf{x}-\mathbf{m}_1)^T C_1^{-1}(\mathbf{x}-\mathbf{m}_1) - (\mathbf{x}-\mathbf{m}_0)^T C_0^{-1}(\mathbf{x}-\mathbf{m}_0)].}$$

For the given covariance matrices,

$$|C_0| = 0.010234375, \qquad |C_1| = 5.112.$$

The inverse class-0 covariance matrix is

$$C_0^{-1} \approx \begin{bmatrix} 2.290076336 & 1.068702290 & -0.152671756 & 0 \\ 1.068702290 & 5.832061069 & 2.595419847 & 0 \\ -0.152671756 & 2.595419847 & 5.343511450 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

The inverse class-1 covariance matrix is

$$C_1^{-1} \approx \begin{bmatrix} 1.120892019 & -0.211267606 & 0.287558685 & 0 \\ -0.211267606 & 0.563380282 & -0.211267606 & 0 \\ 0.287558685 & -0.211267606 & 1.120892019 & 0 \\ 0 & 0 & 0 & 0.333333333 \end{bmatrix}.$$

Because $C_0 \neq C_1$, the quadratic terms do not cancel. The log-likelihood ratio can be written as

$$\ell(\mathbf{x}) = \mathbf{x}^T A\mathbf{x} + \mathbf{b}^T\mathbf{x} + c.$$

The generic coefficients are

$$A = \frac{1}{2}(C_0^{-1} - C_1^{-1}),$$

$$\mathbf{b} = C_1^{-1}\mathbf{m}_1 - C_0^{-1}\mathbf{m}_0,$$

$$c = -\frac{1}{2}(\mathbf{m}_1^T C_1^{-1}\mathbf{m}_1 - \mathbf{m}_0^T C_0^{-1}\mathbf{m}_0) - \frac{1}{2}\ln\frac{|C_1|}{|C_0|}.$$

Numerically,

$$A \approx \begin{bmatrix} 0.584592159 & 0.639984948 & -0.220115221 & 0 \\ 0.639984948 & 2.634340394 & 1.403343726 & 0 \\ -0.220115221 & 1.403343726 & 2.111309716 & 0 \\ 0 & 0 & 0 & 0.833333333 \end{bmatrix},$$

$$\mathbf{b} \approx \begin{bmatrix} 2.800236534 \\ 4.888936673 \\ 5.090312870 \\ 1.333333333 \end{bmatrix},$$

and

$$c \approx -1.7300005201.$$

Therefore the numerical discriminant score is

$$\boxed{\ell(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} - 1.7300005201.}$$

A.2 ROC Curve Using the 10,000 Generated Samples

For a given threshold γ, define

$$D = 1 \quad \text{if} \quad \Lambda(\mathbf{x}) > \gamma,$$

where

$$\Lambda(\mathbf{x}) = \frac{p(\mathbf{x} \mid L = 1)}{p(\mathbf{x} \mid L = 0)}.$$

The true-positive probability is estimated by

$$\hat{P}(D = 1 \mid L = 1; \gamma) = \frac{N_{TP}(\gamma)}{N_1}.$$

The false-positive probability is estimated by

$$\hat{P}(D = 1 \mid L = 0; \gamma) = \frac{N_{FP}(\gamma)}{N_0}.$$

The false-negative probability is

$$\hat{P}(D = 0 \mid L = 1; \gamma) = \frac{N_{FN}(\gamma)}{N_1} = 1 - \hat{P}(D = 1 \mid L = 1; \gamma).$$

The probability-of-error estimate requested by the assignment is

$$\boxed{\hat{P}_e(\gamma) = P(L = 0)\hat{P}(D = 1 \mid L = 0; \gamma) + P(L = 1)\hat{P}(D = 0 \mid L = 1; \gamma).}$$

Substituting the given priors,

$$\hat{P}_e(\gamma) = 0.35\, \hat{P}_{FA}(\gamma) + 0.65\, \hat{P}_{FN}(\gamma).$$

At $\gamma = 0$, all samples are labeled class 1, so

$$(P_{FA}, P_D) = (1,1).$$

As $\gamma \to \infty$, all samples are labeled class 0, so

$$(P_{FA}, P_D) = (0,0).$$

A.3 - Minimum Probability of Error: Empirical Optimum

Searching all finite-sample score intervals gives the minimum empirical error at

$$\boxed{\gamma_{A,\text{emp}} = 0.4453995985}.$$

The corresponding log threshold is

$$\ln\gamma_{A,\text{emp}} = -0.8087834256.$$

At this threshold,

$$N_{TN} = 3378, \qquad N_{FP} = 79, \qquad N_{FN} = 154, \qquad N_{TP} = 6389.$$

The false-positive rate is

$$\hat{P}_{FA} = \frac{79}{3457} = 0.0228521840.$$

The true-positive rate is

$$\hat{P}_D = \frac{6389}{6543} = 0.9764633960.$$

The false-negative rate is

$$\hat{P}_{FN} = \frac{154}{6543} = 0.0235366040.$$

Now substitute into the probability-of-error formula:

$$\hat{P}_e = 0.35(0.0228521840) + 0.65(0.0235366040).$$

$$\hat{P}_e = 0.0079982644 + 0.0152987926.$$

Therefore,

$$\boxed{\hat{P}_{e,A,\text{min}} = 0.0232970570 \approx 2.33\%.}$$

The minimum-error point to mark on the ROC curve is

$$\boxed{(P_{FA}, P_D) = (0.0228521840,\ 0.9764633960).}$$

The finite-sample ROC AUC is

$$\boxed{\mathrm{AUC}_A = 0.9966584953.}$$

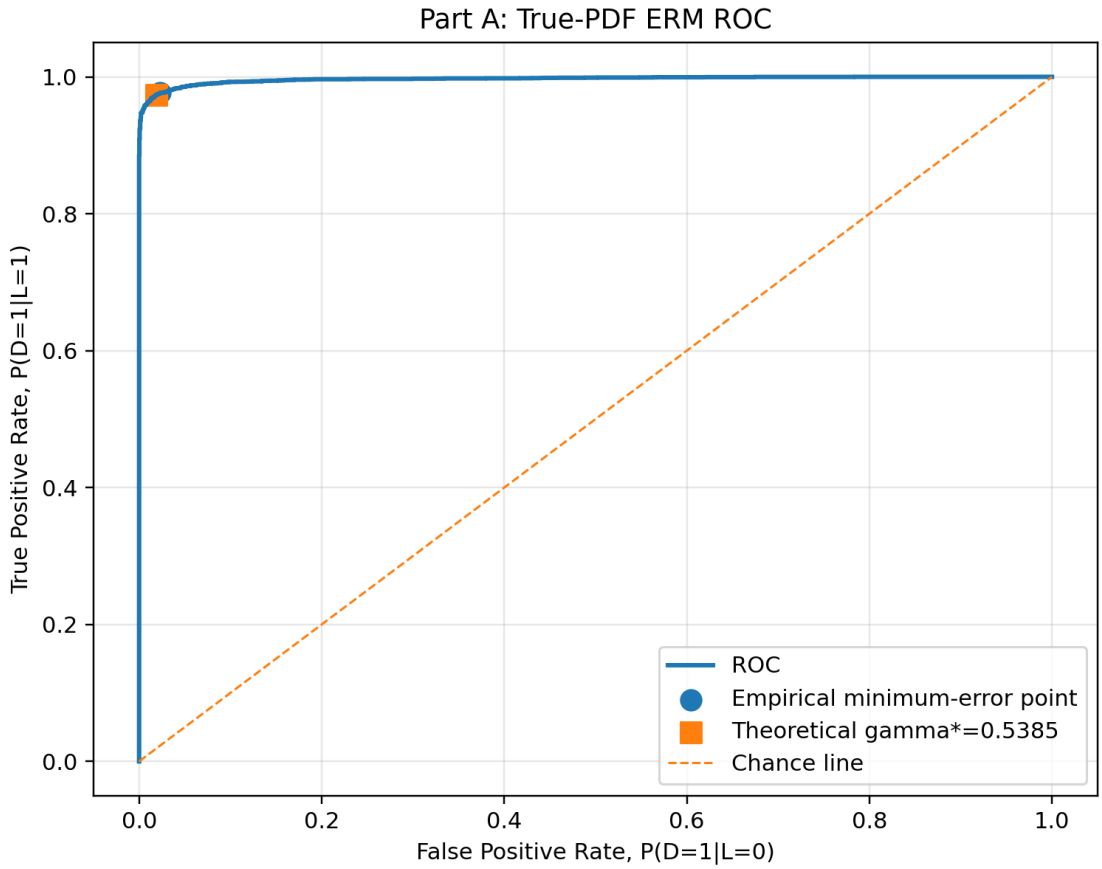


Figure A.1 — ROC curve for the true-PDF ERM classifier; empirical and theoretical operating points are marked.

A.3.1 - Comparison with the Theoretical Threshold

At

$$\gamma^* = 0.5384615385,$$

the confusion counts are

$$N_{TN} = 3391, \qquad N_{FP} = 66, \qquad N_{FN} = 171, \qquad N_{TP} = 6372.$$

Therefore,

$$\hat{P}_{FA} = \frac{66}{3457} = 0.0190916980,$$

$$\hat{P}_D = \frac{6372}{6543} = 0.9738651994,$$

$$\hat{P}_{FN} = \frac{171}{6543} = 0.0261348006.$$

The error estimate is

$$\hat{P}_e = 0.35(0.0190916980) + 0.65(0.0261348006) = 0.0236697147.$$

Thus,

$$\boxed{\hat{P}_e(\gamma^*) \approx 2.37\%.}$$

The empirical optimum 0.4454 differs somewhat from the population-theory threshold 0.53846 because the empirical threshold is selected on one finite random sample. Their error values are nevertheless extremely close: about 2.33% versus 2.37%.

Part B - Naive Bayes Classification with Incorrect Covariance Model

B.1 - Naive Bayes Model and Decision Rule

Conditional independence means that, given the class label,

$$p(\mathbf{x} \mid L = j) = \prod_{r=1}^{4} p\,(x_r \mid L = j).$$

For Gaussian features, this corresponds to using diagonal covariance matrices. The generic diagonal approximation is

$$C_{j,NB} = \mathrm{diag}\big(C_j\big).$$

For class 0,

$$C_{0,NB} = \begin{bmatrix} 0.5 & 0 & 0 & 0 \\ 0 & 0.25 & 0 & 0 \\ 0 & 0 & 0.25 & 0 \\ 0 & 0 & 0 & 0.5 \end{bmatrix}.$$

For class 1,

$$C_{1,NB} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

The true means are retained:

$$\mathbf{m}_{0,NB} = \mathbf{m}_0, \qquad \mathbf{m}_{1,NB} = \mathbf{m}_1.$$

The Naive-Bayes likelihood ratio is

$$\Lambda_{NB}(\mathbf{x}) = \frac{g\big(\mathbf{x} \mid \mathbf{m}_1, C_{1,NB}\big)}{g\big(\mathbf{x} \mid \mathbf{m}_0, C_{0,NB}\big)}.$$

The corresponding log score is

$$\ell_{NB}(\mathbf{x}) = \ln g(\mathbf{x} \mid \mathbf{m}_1, C_{1,NB}) - \ln g(\mathbf{x} \mid \mathbf{m}_0, C_{0,NB}).$$

The 0-1-loss threshold derived from priors and losses is unchanged:

$$\gamma^* = \frac{0.35}{0.65} = 0.5384615385.$$

B.2 - ROC Curve Using the Same 10,000-Sample Dataset

Searching the Naive-Bayes score thresholds gives

$$\boxed{\gamma_{NB,\text{emp}} = 0.3313162736.}$$

Its log-domain value is

$$\ln\gamma_{NB,\text{emp}} = -1.1046818504.$$

At this operating point,

$$N_{TN} = 3336, \qquad N_{FP} = 121, \qquad N_{FN} = 165, \qquad N_{TP} = 6378.$$

Therefore,

$$\hat{P}_{FA} = \frac{121}{3457} = 0.0350014463,$$

$$\hat{P}_{D} = \frac{6378}{6543} = 0.9747822100,$$

and

$$\hat{P}_{FN} = \frac{165}{6543} = 0.0252177900.$$

Substitute into the error expression:

$$\hat{P}_{e,NB} = 0.35(0.0350014463) + 0.65(0.0252177900).$$

$$\hat{P}_{e,NB} = 0.0122505062 + 0.0163915635.$$

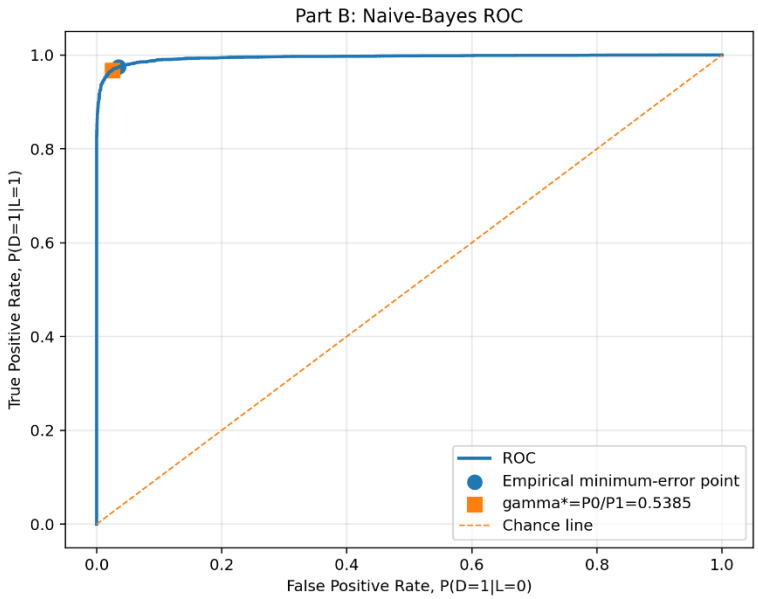
Thus,

$$\boxed{\hat{P}_{e,NB,\min} = 0.0286420697 \approx 2.86\%.}$$

The ROC point to mark is

$$\boxed{(P_{FA}, P_D) = (0.0350014463,\ 0.9747822100).}$$

The finite-sample ROC AUC is



$$\boxed{\mathrm{AUC}_{NB} = 0.9950349595.}$$

Figure B.1 — ROC curve for the Naive Bayes classifier; empirical and prior/loss-based operating points are marked.

B.3 - Minimum Error and Theoretical-Threshold Comparison

At

$$\gamma = 0.5384615385,$$

the confusion counts are

$$N_{TN} = 3370, \qquad N_{FP} = 87, \qquad N_{FN} = 215, \qquad N_{TP} = 6328.$$

Therefore,

$$\hat{P}_{FA} = \frac{87}{3457} = 0.0251663292,$$

$$\hat{P}_{D} = \frac{6328}{6543} = 0.9671404554,$$

$$\hat{P}_{FN} = \frac{215}{6543} = 0.0328595446.$$

The error is

$$\hat{P}_{e,NB}(\gamma^*) = 0.35(0.0251663292) + 0.65(0.0328595446) = 0.0301669192.$$

Hence,

$$\boxed{\hat{P}_{e,NB}(\gamma^*) \approx 3.02\%.}$$

Effect of the Naive Bayes Model Mismatch

The minimum errors are

$$\hat{P}_{e,A,\min} = 0.0232970570,$$

$$\hat{P}_{e,NB,\min} = 0.0286420697.$$

The absolute increase in error probability is

$$0.0286420697 - 0.0232970570 = 0.0053450127.$$

Thus the absolute increase is about

$$\boxed{0.535 \text{ percentage points.}}$$

The relative increase is

$$\frac{0.0286420697 - 0.0232970570}{0.0232970570} \times 100\% \approx 22.94\%.$$

Therefore, the incorrect conditional-independence assumption negatively affects classification performance. The effect is visible both in the larger minimum error and in the slight AUC reduction from 0.99666 to 0.99503.

Part C Fisher Linear Discriminant Analysis

C.1 - Estimate Means, Covariances, and Fisher Projection

For class j with N_j samples, the sample mean estimator is

$$\boxed{\hat{\mathbf{m}}_j = \frac{1}{N_j} \sum_{i:L_i=j} \mathbf{x}_i .}$$

The unbiased sample covariance estimator is

$$\boxed{\hat{C}_j = \frac{1}{N_j - 1} \sum_{i:L_i=j} \left(\mathbf{x}_i - \hat{\mathbf{m}}_j\right)\left(\mathbf{x}_i - \hat{\mathbf{m}}_j\right)^T .}$$

Applying these formulas to the same 10,000-sample data set gives

$$\hat{\mathbf{m}}_0 \approx \begin{bmatrix} -0.531499550 \\ -0.491948820 \\ -0.503583327 \\ -0.493724106 \end{bmatrix},$$

$$\hat{\mathbf{m}}_1 \approx \begin{bmatrix} 0.995777638 \\ 1.023206594 \\ 1.013990770 \\ 1.005926654 \end{bmatrix}.$$

The estimated class-0 covariance is

$$\hat{C}_0 \approx \begin{bmatrix} 0.487624643 & -0.121613126 & 0.069389199 & -0.010429844 \\ -0.121613126 & 0.250548766 & -0.132883770 & 0.007373093 \\ 0.069389199 & -0.132883770 & 0.257473006 & -0.008908443 \\ -0.010429844 & 0.007373093 & -0.008908443 & 0.502840911 \end{bmatrix}.$$

The estimated class-1 covariance is

$$\hat{C}_1 \approx \begin{bmatrix} 1.019010268 & 0.320431023 & -0.202358779 & 0.013866668 \\ 0.320431023 & 2.006201626 & 0.301717759 & 0.067090473 \\ -0.202358779 & 0.301717759 & 1.000299218 & -0.002929099 \\ 0.013866668 & 0.067090473 & -0.002929099 & 3.030783136 \end{bmatrix}.$$

C.1.1 - Within-Class and Between-Class Scatter Matrices

The assignment specifies equal class weighting when constructing the Fisher scatter matrices. The equally weighted mean is therefore

$$\boxed{\hat{\mathbf{m}} = \frac{1}{2}\hat{\mathbf{m}}_0 + \frac{1}{2}\hat{\mathbf{m}}_1.}$$

Numerically,

$$\hat{\mathbf{m}} \approx \begin{bmatrix} 0.232139044 \\ 0.265628887 \\ 0.255203721 \\ 0.256101274 \end{bmatrix}.$$

Using equal covariance weights, the within-class scatter matrix is

$$\boxed{S_W = \frac{1}{2}\hat{C}_0 + \frac{1}{2}\hat{C}_1.}$$

Numerically,

$$S_W \approx \begin{bmatrix} 0.753317456 & 0.099408949 & -0.066484790 & 0.001718412 \\ 0.099408949 & 1.128375196 & 0.084416994 & 0.037231783 \\ -0.066484790 & 0.084416994 & 0.628886112 & -0.005918771 \\ 0.001718412 & 0.037231783 & -0.005918771 & 1.766812023 \end{bmatrix}.$$

The generic between-class scatter matrix with equal class weights is

$$\boxed{S_B = \frac{1}{2}(\hat{\mathbf{m}}_0 - \hat{\mathbf{m}})(\hat{\mathbf{m}}_0 - \hat{\mathbf{m}})^T + \frac{1}{2}(\hat{\mathbf{m}}_1 - \hat{\mathbf{m}})(\hat{\mathbf{m}}_1 - \hat{\mathbf{m}})^T.}$$

Numerically,

$$S_B \approx \begin{bmatrix} 0.583143902 & 0.578515575 & 0.579439075 & 0.572595599 \\ 0.578515575 & 0.573923982 & 0.574840152 & 0.568050992 \\ 0.579439075 & 0.574840152 & 0.575757785 & 0.568957787 \\ 0.572595599 & 0.568050992 & 0.568957787 & 0.562238100 \end{bmatrix}.$$

C.1.2 - Fisher Criterion and Generalized Eigenvalue Problem

The generic Fisher objective is

$$\boxed{J(\mathbf{w}) = \frac{\mathbf{w}^T S_B \mathbf{w}}{\mathbf{w}^T S_W \mathbf{w}}.}$$

Maximizing this Rayleigh quotient produces the generalized eigenvalue problem

$$\boxed{S_B \mathbf{w} = \lambda S_W \mathbf{w}.}$$

The eigenvector associated with the largest generalized eigenvalue is the Fisher projection direction. For this data set,

$$\lambda_{\max} = 2.4358700112.$$

After choosing the sign so that class 1 has the larger projected mean and normalizing to unit Euclidean norm,

$$\boxed{\mathbf{w}_{LDA} \approx \begin{bmatrix} 0.602193729 \\ 0.266418599 \\ 0.714340548 \\ 0.236857391 \end{bmatrix}.}$$

The projected sample means are

$$\mathbf{w}_{LDA}^T \hat{\mathbf{m}}_0 = -0.9278022049,$$

$$\mathbf{w}_{LDA}^T \hat{\mathbf{m}}_1 = 1.8348482024.$$

Therefore class 1 lies on the larger-score side of the projection.

C.2 - LDA Threshold Sweep and ROC Curve

The scalar Fisher score is

$$y = \mathbf{w}_{LDA}^T \mathbf{x}.$$

The requested threshold decision rule is

$$D(\mathbf{x}) = \begin{cases} 1, & \mathbf{w}_{LDA}^T \mathbf{x} > \tau, \\ 0, & \mathbf{w}_{LDA}^T \mathbf{x} \leq \tau. \end{cases}$$

Varying τ from $-\infty$ to $+\infty$ produces the LDA ROC curve.

C.3 - Minimum Probability of Error and Operating Point

The minimum empirical error occurs at

$$\boxed{\tau_{\min} = -0.0498128006.}$$

At this threshold,

$$N_{TN} = 3277, \qquad N_{FP} = 180, \qquad N_{FN} = 291, \qquad N_{TP} = 6252.$$

Therefore,

$$\hat{P}_{FA} = \frac{180}{3457} = 0.0520682673,$$

$$\hat{P}_{D} = \frac{6252}{6543} = 0.9555249885,$$

and

$$\hat{P}_{FN} = \frac{291}{6543} = 0.0444750115.$$

Substitute into the error formula:

$$\hat{P}_{e,LDA} = 0.35(0.0520682673) + 0.65(0.0444750115).$$

$$\hat{P}_{e,LDA} = 0.0182238935 + 0.0289087575.$$

Thus,

$$\boxed{\hat{P}_{e,LDA,\min} = 0.0471326510 \approx 4.71\%.}$$

The ROC point to mark is

$$\boxed{(P_{FA}, P_D) = (0.0520682673,\ 0.9555249885).}$$

The finite-sample ROC AUC is

$$\boxed{\mathrm{AUC}_{LDA} = 0.9858060543.}$$

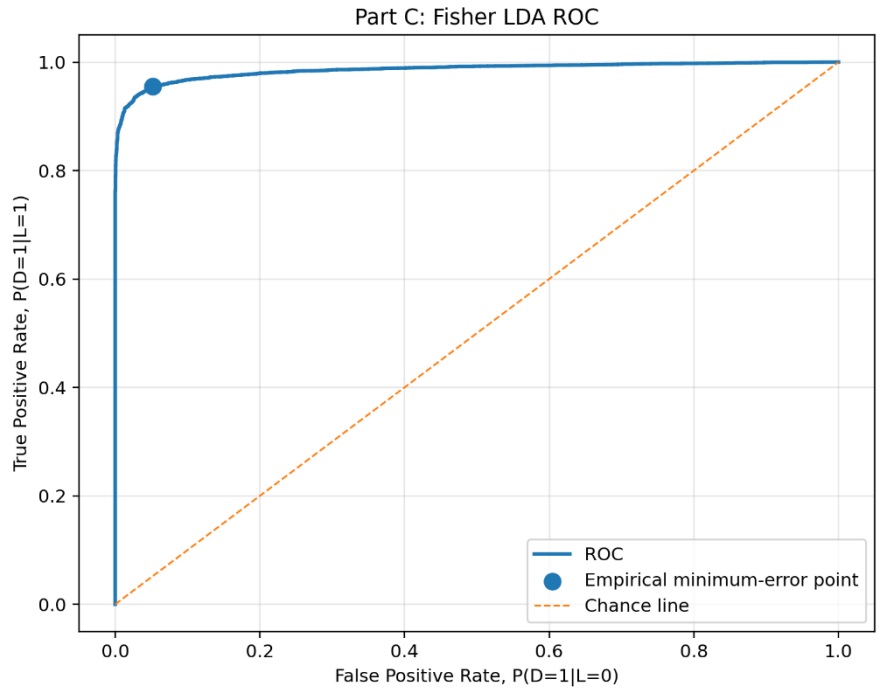


Figure C.1 — ROC curve for the Fisher LDA classifier with the empirical minimum-error operating point marked.

Overall Comparison of Parts A, B, and C

Case	Threshold	FPR	TPR	P(error)	AUC
A - empirical	γ = 0.445400	0.022852	0.976463	**0.023297**	**0.996658**
A - theoretical	γ* = 0.538462	0.019092	0.973865	0.023670	0.996658
B - empirical	γ = 0.331316	0.035001	0.974782	0.028642	0.995035
B - prior/loss threshold	γ = 0.538462	0.025166	0.967140	0.030167	0.995035
C - empirical	τ = -0.049813	0.052068	0.955525	0.047133	0.985806

The table compares the empirical minimum-error operating points for Parts A, B, and C. The additional theoretical/prior-loss rows show how the fixed 0-1-loss threshold performs for Parts A and B on the same finite 10,000-sample data set.

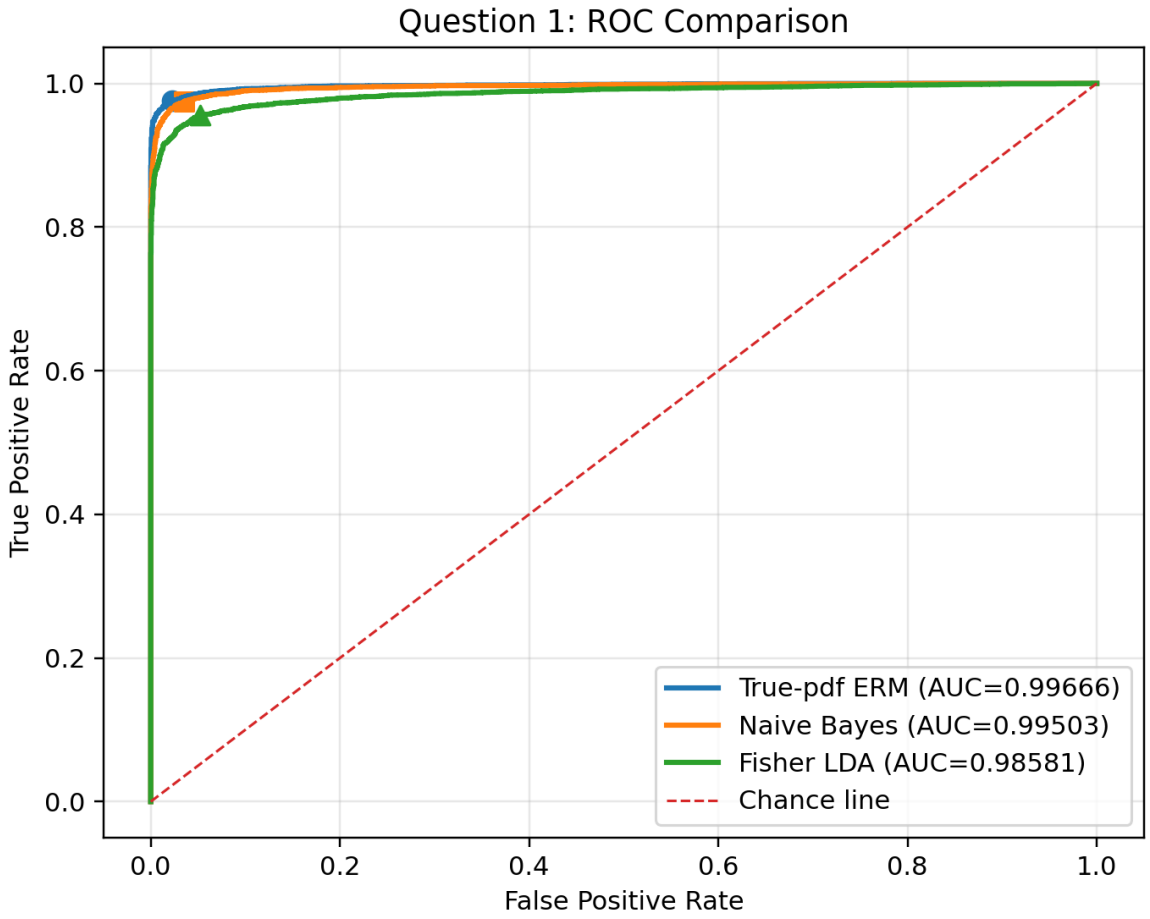


Figure 1 — ROC comparison for the true-PDF ERM, Naive Bayes, and Fisher LDA classifiers.

The ordering of performance is

$$\boxed{\text{True-PDF ERM} > \text{Naive Bayes} > \text{Fisher LDA.}}$$

The true-PDF ERM classifier performs best because it uses the correct class-conditional Gaussian distributions, including their full covariance matrices. Since $C_0 \neq C_1$, the optimal log-likelihood ratio contains quadratic terms and produces a quadratic decision boundary.

Naive Bayes removes the nonzero cross-covariances. This model mismatch loses information about feature dependence and increases the minimum observed probability of error from approximately 2.33% to 2.86%.

Fisher LDA compresses the four-dimensional feature vector to one linear score. This is less flexible than the quadratic true-PDF rule, which explains the larger minimum error of approximately 4.71% for this experiment.

Answer 2:

Problem setup and selected Gaussian model

The problem requires a 3-dimensional random vector $\mathbf{X} \in \mathbb{R}^3$ generated from four Gaussian components. Class 1 is represented by one Gaussian, class 2 by one Gaussian, and class 3 by an equal-weight mixture of the remaining two Gaussian components. The required class priors are

$$P(L = 1) = 0.3, \qquad P(L = 2) = 0.3, \qquad P(L = 3) = 0.4.$$

For a 3-dimensional Gaussian, the probability density is

$$g(\mathbf{x} \mid \boldsymbol{\mu}, \mathbf{C}) = \frac{1}{(2\pi)^{3/2}|\mathbf{C}|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{C}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right].$$

The following four Gaussian components were selected so that their mean separations are approximately 2 to 3 times a representative component standard deviation, which creates meaningful overlap between classes.

Class 1

$$p(\mathbf{x} \mid L = 1) = g(\mathbf{x} \mid \boldsymbol{\mu}_1, \mathbf{C}_1),$$

$$\boldsymbol{\mu}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \qquad \mathbf{C}_1 = \begin{bmatrix} 1.0 & 0.2 & 0.1 \\ 0.2 & 0.8 & 0.1 \\ 0.1 & 0.1 & 0.9 \end{bmatrix}.$$

Class 2

$$p(\mathbf{x} \mid L = 2) = g(\mathbf{x} \mid \boldsymbol{\mu}_2, \mathbf{C}_2),$$

$$\boldsymbol{\mu}_2 = \begin{bmatrix} 2.4 \\ 0.4 \\ 0 \end{bmatrix}, \qquad \mathbf{C}_2 = \begin{bmatrix} 0.9 & -0.2 & 0 \\ -0.2 & 1.0 & 0.15 \\ 0 & 0.15 & 0.8 \end{bmatrix}.$$

Class 3

Class 3 is an equal-weight mixture of two Gaussian components:

$$p(\mathbf{x} \mid L = 3) = \frac{1}{2} g(\mathbf{x} \mid \boldsymbol{\mu}_{3a}, \mathbf{C}_{3a}) + \frac{1}{2} g(\mathbf{x} \mid \boldsymbol{\mu}_{3b}, \mathbf{C}_{3b}).$$

$$\boldsymbol{\mu}_{3a} = \begin{bmatrix} 1.0 \\ 2.2 \\ 0.3 \end{bmatrix}, \qquad \mathbf{C}_{3a} = \begin{bmatrix} 0.8 & 0.1 & -0.1 \\ 0.1 & 0.9 & 0.2 \\ -0.1 & 0.2 & 1.0 \end{bmatrix},$$

$$\boldsymbol{\mu}_{3b} = \begin{bmatrix} 1.2 \\ 0.8 \\ 2.2 \end{bmatrix}, \qquad \mathbf{C}_{3b} = \begin{bmatrix} 1.0 & 0.15 & 0.05 \\ 0.15 & 0.8 & -0.15 \\ 0.05 & -0.15 & 0.9 \end{bmatrix}.$$

Check of the requested Gaussian overlap

A convenient representative standard deviation is the average of the 12 marginal standard deviations from the four covariance matrices:

$$\sigma_{\text{avg}} = \frac{1}{12}\sum_{k=1}^{4}\sum_{d=1}^{3}\sqrt{[\mathbf{C}_k]_{dd}} = 0.9477.$$

The pairwise Euclidean distances between component means are shown below.

Pairwise mean-separation check

μ1 to μ2: distance = 2.4331; distance / sigma_avg = 2.5674.

μ1 to μ3a: distance = 2.4352; distance / sigma_avg = 2.5695.

μ1 to μ3b: distance = 2.6306; distance / sigma_avg = 2.7758.

μ2 to μ3a: distance = 2.3000; distance / sigma_avg = 2.4269.

μ2 to μ3b: distance = 2.5377; distance / sigma_avg = 2.6778.

μ3a to μ3b: distance = 2.3685; distance / sigma_avg = 2.4992.

Thus, every pair is separated by approximately $2.43\sigma_{\text{avg}}$ to $2.78\sigma_{\text{avg}}$, satisfying the requested approximate 2-to-3-standard-deviation separation while preserving significant overlap.

I have divided the answer into two parts (Part A and Part B)

Part A - Minimum Probability of Error Classification (0-1 Loss / MAP)

A.1 Generate 10,000 samples and keep the true labels

A single dataset of

$$N = 10{,}000$$

samples was generated using random seed 42 so that the numerical results are reproducible. First, a class label L_n was sampled according to the class priors. Conditional on that label:

- if $L_n = 1$, the sample was drawn from $g(\mathbf{x} \mid \mu_1, C_1)$;
- if $L_n = 2$, the sample was drawn from $g(\mathbf{x} \mid \mu_2, C_2)$;
- if $L_n = 3$, one of the two class-3 Gaussian components was selected with probability 0.5 and the sample was drawn from that component.

The resulting class counts were

$$N_1 = 3043, \qquad N_2 = 3005, \qquad N_3 = 3952,$$

which sum to 10,000 and are close to the expected proportions 0.3, 0.3, and 0.4. Inside class 3, 1953 samples came from component $3a$ and 1999 from component $3b$.

A.2 Specify and implement the minimum-error decision rule

With 0-1 loss, the Bayes decision rule minimizes the probability of classification error. The optimal decision is the maximum-a-posteriori (MAP) rule:

$$D_{\text{MAP}}(\mathbf{x}) = \arg\max_{j\in\{1,2,3\}} P(L = j \mid \mathbf{x}).$$

Bayes' rule gives

$$P(L = j \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid L = j)P(L = j)}{p(\mathbf{x})}.$$

Because $p(\mathbf{x})$ is common to every class, it can be omitted from the comparison. Define the joint class scores

$$s_j(\mathbf{x}) = P(L = j)p(\mathbf{x} \mid L = j).$$

For the selected model,

$$s_1(\mathbf{x}) = 0.3\, g(\mathbf{x} \mid \mu_1, C_1),$$
$$s_2(\mathbf{x}) = 0.3\, g(\mathbf{x} \mid \mu_2, C_2),$$

and

$$s_3(\mathbf{x}) = 0.4\left[\frac{1}{2}g(\mathbf{x} \mid \mu_{3a}, C_{3a}) + \frac{1}{2}g(\mathbf{x} \mid \mu_{3b}, C_{3b})\right].$$

Therefore, the implemented classifier is

$$\boxed{D_{\text{MAP}}(\mathbf{x}) = \operatorname{argmax}\{s_1(\mathbf{x}), s_2(\mathbf{x}), s_3(\mathbf{x})\}.}$$

Empirical confusion matrix

The problem defines confusion-matrix entries as

$$P(D = i \mid L = j),$$

so **rows correspond to the decision** $D = i$ **and columns correspond to the true class** $L = j$.

The sample-count matrix was

$$\mathbf{C}_{\text{MAP}} = \begin{bmatrix} 2454 & 205 & 314 \\ 217 & 2409 & 276 \\ 372 & 391 & 3362 \end{bmatrix}.$$

Column normalization produces the requested empirical conditional confusion matrix:

$$\hat{P}(D = i \mid L = j) = \begin{bmatrix} 0.80644 & 0.06822 & 0.07945 \\ 0.07131 & 0.80166 & 0.06984 \\ 0.12225 & 0.13012 & 0.85071 \end{bmatrix}.$$

For example, for true class 1,

$$\hat{P}(D = 1 \mid L = 1) = \frac{2454}{3043} = 0.80644,$$
$$\hat{P}(D = 2 \mid L = 1) = \frac{217}{3043} = 0.07131,$$
$$\hat{P}(D = 3 \mid L = 1) = \frac{372}{3043} = 0.12225.$$

The three probabilities sum to 1, as required. The corresponding correct-classification probabilities are approximately 80.64% for class 1, 80.17% for class 2, and 85.07% for class 3.

Empirical probability of error

The number of incorrect MAP decisions is

$$N_{\text{error}} = 217 + 372 + 205 + 391 + 314 + 276 = 1775.$$

Therefore,

$$\boxed{\hat{P}_{\text{error,MAP}} = \frac{1775}{10000} = 0.1775 = 17.75\%.}$$

Using the known priors together with the estimated conditional confusion probabilities gives

$$\begin{aligned}\hat{P}_e &= 0.3[1 - P(D = 1 \mid L = 1)] + 0.3[1 - P(D = 2 \mid L = 2)] + 0.4[1 - P(D = 3 \mid L = 3)] \\ &\approx 0.17729.\end{aligned}$$

The small difference between 0.17729 and the direct sample fraction 0.1775 is due to the generated class proportions being close to, but not exactly, 0.3, 0.3, and 0.4.

A.3 Required 3D visualization

The plot in the next page uses marker shape for the **true class label** and marker color for classification correctness:

- dot: true class 1;
- circle: true class 2;
- triangle: true class 3;
- green: correctly classified;
- red: incorrectly classified.

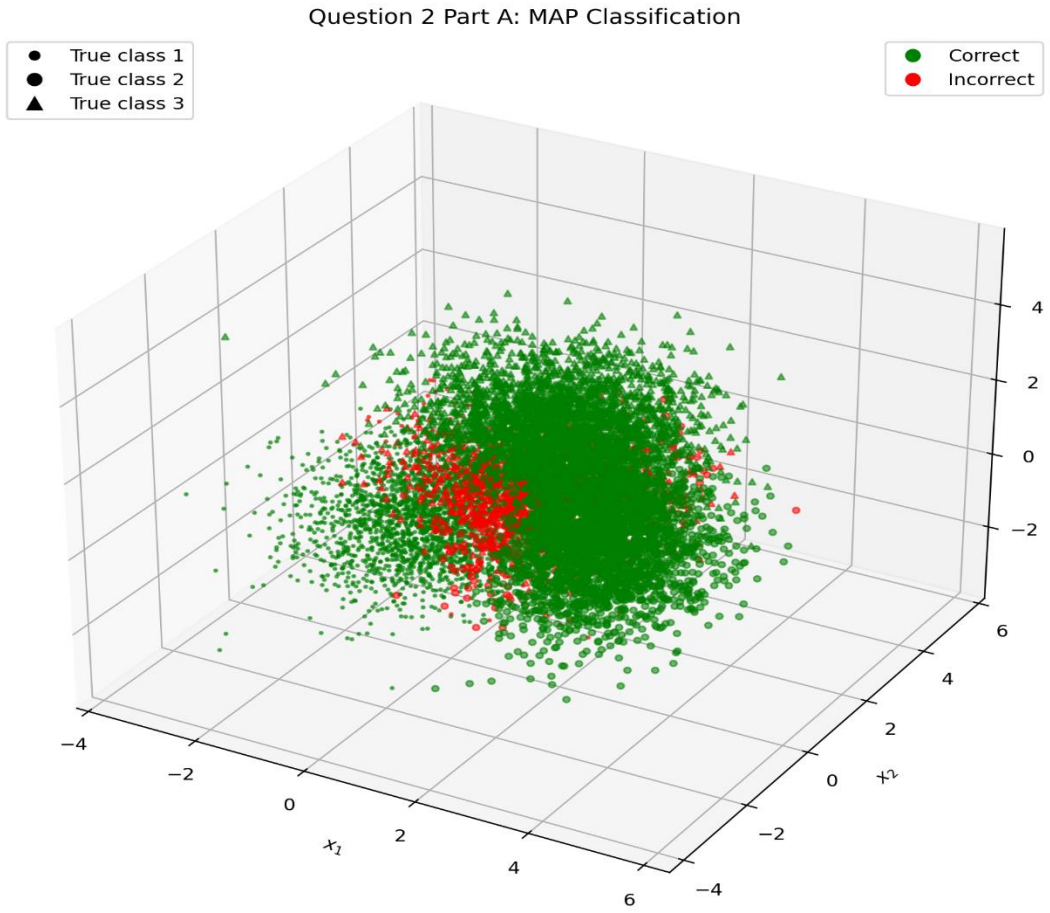


Figure 1. Three-dimensional data visualization for the MAP classifier. Marker shape denotes the true class; green denotes a correct decision and red denotes an error.

The red points concentrate primarily in regions where the Gaussian class-conditional densities overlap. This is expected because the Gaussian means were deliberately chosen to create significant overlap. Under 0-1 loss, the MAP rule is nevertheless optimal for minimizing the overall probability of error for the specified true distribution.

Part B - Minimum Expected Risk (ERM) Classification

The loss is now asymmetric. For a general loss matrix $\Lambda = [\lambda_{ij}]$, where λ_{ij} is the loss incurred by deciding class i when the true class is j, the conditional risk of decision i is

$$R(i \mid \mathbf{x}) = \sum_{j=1}^{3} \lambda_{ij} P(L = j \mid \mathbf{x}).$$

The Bayes minimum-expected-risk classifier is therefore

$$\boxed{D_{\text{ERM}}(\mathbf{x}) = \arg\min_{i\in\{1,2,3\}} R(i \mid \mathbf{x}).}$$

The posterior probabilities can be evaluated from the same true data model used in Part A:

$$P(L = j \mid \mathbf{x}) = \frac{s_j(\mathbf{x})}{s_1(\mathbf{x}) + s_2(\mathbf{x}) + s_3(\mathbf{x})}.$$

Since the denominator is positive and common to all decisions, it does not change which risk is smallest.

B.1 ERM classifier for Λ_{10}

The first loss matrix is

$$\Lambda_{10} = \begin{bmatrix} 0 & 1 & 10 \\ 1 & 0 & 10 \\ 1 & 1 & 0 \end{bmatrix}.$$

The three conditional risks are

$$R_1 = P(L = 2 \mid \mathbf{x}) + 10P(L = 3 \mid \mathbf{x}),$$
$$R_2 = P(L = 1 \mid \mathbf{x}) + 10P(L = 3 \mid \mathbf{x}),$$
$$R_3 = P(L = 1 \mid \mathbf{x}) + P(L = 2 \mid \mathbf{x}).$$

Using the joint scores $s_j(\mathbf{x})$, the equivalent unnormalized quantities are

$$\tilde{R}_1 = s_2 + 10s_3, \qquad \tilde{R}_2 = s_1 + 10s_3, \qquad \tilde{R}_3 = s_1 + s_2.$$

Pairwise comparison shows that decision 1 requires $s_1 > s_2$ and $s_1 > 10s_3$, decision 2 requires $s_2 > s_1$ and $s_2 > 10s_3$, and otherwise decision 3 is optimal. Hence an equivalent convenient rule is

$$\boxed{D_{\Lambda_{10}}(\mathbf{x}) = \text{argmax}\{s_1(\mathbf{x}), s_2(\mathbf{x}), 10s_3(\mathbf{x})\}.}$$

This form makes the effect of the loss matrix intuitive: because missing a true class-3 sample is ten times as costly, the class-3 score is effectively emphasized by a factor of 10 in the decision comparison.

Numerical results for Λ_{10}

The count matrix is

$$\mathbf{C}_{10} = \begin{bmatrix} 1647 & 75 & 41 \\ 85 & 1652 & 32 \\ 1311 & 1278 & 3879 \end{bmatrix}.$$

The conditional confusion matrix is

$$\hat{P}_{10}(D = i \mid L = j) = \begin{bmatrix} 0.54124 & 0.02496 & 0.01037 \\ 0.02793 & 0.54975 & 0.00810 \\ 0.43082 & 0.42529 & 0.98153 \end{bmatrix}.$$

The probability of correctly recognizing a true class-3 sample rises to

$$P(D = 3 \mid L = 3) = \frac{3879}{3952} = 0.98153 = 98.153\%.$$

Only

$$41 + 32 = 73$$

true class-3 samples are assigned to classes 1 or 2.

Estimated minimum expected risk for Λ_{10}

From the conditional confusion probabilities and the known class priors,

$$\hat{R} = \sum_{j=1}^{3} P\,(L = j) \sum_{i=1}^{3} \lambda_{ij}\, \hat{P}(D = i \mid L = j).$$

For class 1, every wrong decision has unit loss, so

$$\hat{R}_{10|L=1} = \frac{85 + 1311}{3043} = 0.45876.$$

For class 2,

$$\hat{R}_{10|L=2} = \frac{75 + 1278}{3005} = 0.45025.$$

For class 3, an error has loss 10:

$$\hat{R}_{10|L=3} = 10\left(\frac{41 + 32}{3952}\right) = 0.18472.$$

Therefore,

$$\hat{R}_{10} = 0.3(0.45876) + 0.3(0.45025) + 0.4(0.18472) = \boxed{0.34659}.$$

As a direct sample-average check,

$$\hat{R}_{10,\text{sample}} = \frac{1}{10000} \sum_{n=1}^{10000} \lambda_{D_n, L_n} = 0.3479.$$

The ordinary error probability is

$$\hat{P}_{e,10} = \frac{2822}{10000} = \boxed{0.2822 = 28.22\%}.$$

This error probability is higher than MAP because the ERM rule is no longer minimizing the number of errors; it is minimizing their asymmetric cost.

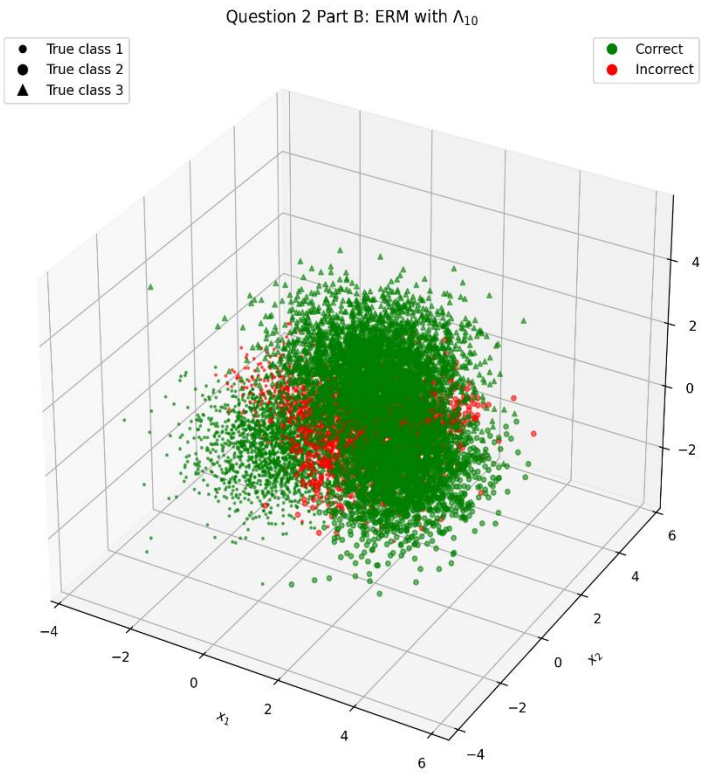


Figure 2. Three-dimensional classification visualization for the ERM rule using Λ_{10}.

The figure shows a visibly larger class-3 decision region. As a result, almost all true class-3 samples are protected, while more class-1 and class-2 samples are intentionally sacrificed by assigning them to class 3.

B.2 ERM classifier for Λ_{100}

The second loss matrix is

$$\Lambda_{100} = \begin{bmatrix} 0 & 1 & 100 \\ 1 & 0 & 100 \\ 1 & 1 & 0 \end{bmatrix}.$$

The conditional risks become

$$R_1 = P(L = 2 \mid \mathbf{x}) + 100P(L = 3 \mid \mathbf{x}),$$
$$R_2 = P(L = 1 \mid \mathbf{x}) + 100P(L = 3 \mid \mathbf{x}),$$
$$R_3 = P(L = 1 \mid \mathbf{x}) + P(L = 2 \mid \mathbf{x}).$$

The equivalent score comparison is

$$\boxed{D_{\Lambda_{100}}(\mathbf{x}) = \operatorname{argmax}\{s_1(\mathbf{x}), s_2(\mathbf{x}), 100s_3(\mathbf{x})\}.}$$

Thus, the classifier requires extremely strong evidence for class 1 or class 2 before it is willing to risk deciding against class 3.

Numerical results for Λ_{100}

The count matrix is

$$\mathbf{C}_{100} = \begin{bmatrix} 697 & 10 & 0 \\ 10 & 803 & 1 \\ 2336 & 2192 & 3951 \end{bmatrix}.$$

The conditional confusion matrix is

$$\hat{P}_{100}(D = i \mid L = j) = \begin{bmatrix} 0.22905 & 0.00333 & 0 \\ 0.00329 & 0.26722 & 0.000253 \\ 0.76766 & 0.72945 & 0.999747 \end{bmatrix}.$$

For true class 3,

$$P(D = 3 \mid L = 3) = \frac{3951}{3952} = 0.999747 = 99.9747\%.$$

Only one of the 3952 class-3 samples is misclassified.

Estimated minimum expected risk for Λ_{100}

For class 1,

$$\hat{R}_{100|L=1} = \frac{10 + 2336}{3043} = 0.77095.$$

For class 2,

$$\hat{R}_{100|L=2} = \frac{10 + 2192}{3005} = 0.73278.$$

For class 3, a mistake has loss 100:

$$\hat{R}_{100|L=3} = 100\left(\frac{0 + 1}{3952}\right) = 0.025304.$$

Therefore,

$$\hat{R}_{100} = 0.3(0.77095) + 0.3(0.73278) + 0.4(0.025304) = \boxed{0.46124}.$$

The direct sample-average loss is

$$\hat{R}_{100,\text{sample}} = 0.4648.$$

The ordinary error probability is

$$\hat{P}_{e,100} = \frac{4549}{10000} = \boxed{0.4549 = 45.49\%}.$$

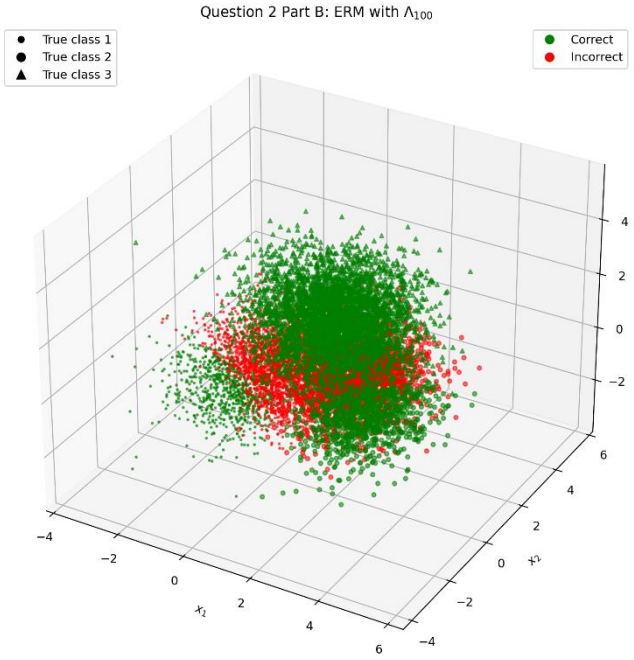


Figure 3. Three-dimensional classification visualization for the ERM rule using Λ_{100}.

The plot confirms the strong shift toward class 3: nearly every true class-3 sample is classified correctly, but a large fraction of classes 1 and 2 are deliberately assigned to class 3 to avoid the very large cost associated with missing class 3.

Comparison

The main numerical results are summarized below.

MAP / 0-1 loss: class-1 correct rate = 0.8064; class-2 correct rate = 0.8017; class-3 correct rate = 0.8507; ordinary error probability = 0.1775; prior-weighted estimated risk = 0.1773.

ERM with Lambda10: class-1 correct rate = 0.5412; class-2 correct rate = 0.5498; class-3 correct rate = 0.9815; ordinary error probability = 0.2822; estimated expected risk = 0.3466.

ERM with Lambda100: class-1 correct rate = 0.2291; class-2 correct rate = 0.2672; class-3 correct rate = 0.99975; ordinary error probability = 0.4549; estimated expected risk = 0.4612.

Under 0-1 loss, every classification error has the same cost, so the MAP classifier minimizes the total error probability and achieves the smallest ordinary error rate, approximately 17.75%.

Under Λ_{10}, an error when the true class is 3 costs ten times more than the ordinary class-1/class-2 mistakes. The optimal ERM rule therefore expands the effective class-3 decision region. The true-

class-3 accuracy increases from 85.07% to 98.15%, while ordinary error increases to 28.22% because many class-1 and class-2 samples are now assigned to class 3.

Under Λ_{100}, protecting class 3 is even more important. The classifier correctly identifies 3951 of 3952 class-3 samples, or 99.9747%. The cost is a substantial increase in ordinary error to 45.49%, mainly because 76.77% of true class-1 samples and 72.95% of true class-2 samples are classified as class 3.

This is not a contradiction. The three classifiers optimize different objectives. MAP minimizes

$$P(D \neq L),$$

whereas the asymmetric ERM classifiers minimize

$$E[\lambda_{D,L}].$$

A useful verification is to evaluate the MAP decisions under the asymmetric losses. Using Λ_{10}, the MAP rule would have estimated risk approximately 0.7147, while the optimal Λ_{10} ERM rule reduces it to 0.3466. Using Λ_{100}, MAP would have risk approximately 6.0892, while the corresponding ERM rule reduces it to 0.4612. Therefore, the higher ordinary error rates in Part B are the correct consequence of minimizing asymmetric expected loss rather than simple error count.

The numerical risks 0.3466 and 0.4612 should not be compared directly to decide which loss matrix is "better," because the two loss matrices use different numerical cost scales. The proper comparison is between competing decision rules evaluated under the same loss matrix.

For Part A, the MAP classifier is

$$\boxed{D_{\text{MAP}}(\mathbf{x}) = \underset{j}{\operatorname{argmax}} P(L = j)p(\mathbf{x} \mid L = j)}$$

with empirical confusion matrix

$$\boxed{\begin{bmatrix} 0.80644 & 0.06822 & 0.07945 \\ 0.07131 & 0.80166 & 0.06984 \\ 0.12225 & 0.13012 & 0.85071 \end{bmatrix}}$$

and empirical probability of error

$$\boxed{\hat{P}_e = 0.1775.}$$

For Λ_{10},

$$\boxed{\hat{R}_{10} = 0.34659, \qquad \hat{P}_{e,10} = 0.2822.}$$

For Λ_{100},

$$\boxed{\hat{R}_{100} = 0.46124, \qquad \hat{P}_{e,100} = 0.4549.}$$

Answer 3:

1. Modeling Approach and Assumptions

For each class encountered in a dataset, the feature vector was modeled by a separate multivariate Gaussian class-conditional density. A different mean vector and a full covariance matrix were estimated for every class. Class priors were estimated directly from class sample counts.

$$p(\mathbf{x} \mid Y = k) = \frac{1}{(2\pi)^{d/2}|\mathbf{C}_k|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \mathbf{C}_k^{-1}(\mathbf{x} - \boldsymbol{\mu}_k)\right]$$

The White Wine dataset contains 4,898 samples and 11 features. The encountered quality classes are 3 through 9. For Human Activity Recognition (HAR), the original UCI train and test partitions were combined to use all 10,299 samples and all 561 features, as required by the assignment.

1.1 Parameter Estimation

For class k containing N_k samples out of N total samples, the prior was estimated as:

$$\hat{\pi}_k = \hat{P}(Y = k) = \frac{N_k}{N}$$

The class mean vector was estimated using the sample average:

$$\hat{\boldsymbol{\mu}}_k = \frac{1}{N_k} \sum_{n\,:\,y_n = k} \mathbf{x}_n$$

The class covariance matrix was estimated using the sample-average / maximum-likelihood form:

$$\hat{\mathbf{C}}_k = \frac{1}{N_k} \sum_{n\,:\,y_n = k} (\mathbf{x}_n - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_n - \hat{\boldsymbol{\mu}}_k)^T$$

1.2 Covariance Regularization

Some covariance estimates are singular or ill-conditioned, especially for classes with few samples and for the high-dimensional HAR dataset. A diagonal regularization term was therefore added. The regularization scale was chosen using the arithmetic mean of the nonzero covariance eigenvalues, with α = 0.001.

$$\mathbf{C}_{k,\,\text{reg}} = \hat{\mathbf{C}}_k + \lambda_k \mathbf{I}, \qquad \lambda_k = \alpha \frac{\text{trace}(\hat{\mathbf{C}}_k)}{\text{rank}(\hat{\mathbf{C}}_k)}, \qquad \alpha = 10^{-3}$$

This guarantees a positive shift of the covariance eigenvalues and makes the Gaussian discriminant numerically stable. The Wine quality-9 covariance has rank 4 because only five samples are available. For HAR, the covariance ranks are approximately 466-469, below the feature dimension 561, so regularization is essential.

1.3 Minimum-Probability-of-Error Decision Rule

Under 0-1 loss, the minimum-probability-of-error rule is the MAP rule. After taking the logarithm of the Gaussian posterior and dropping terms common to all classes, the discriminant used for class k is:

$$g_k(\mathbf{x}) = \log \hat{\pi}_k - \tfrac{1}{2}\log|\mathbf{C}_{k,\text{reg}}| - \tfrac{1}{2}(\mathbf{x} - \hat{\boldsymbol{\mu}}_k)^T \mathbf{C}_{k,\text{reg}}^{-1}(\mathbf{x} - \hat{\boldsymbol{\mu}}_k)$$

The predicted class is:

$$\hat{y} = \arg\max_k g_k(\mathbf{x})$$

The empirical training error probability is:

$$\hat{P}_e = \frac{N_{\text{error}}}{N}$$

All available samples were used to estimate the Gaussian models and the same samples were then classified, exactly as requested. Therefore, the reported error probabilities are training (resubstitution) error estimates rather than independent test-set errors.

2. Wine Quality Dataset

The white-wine data contain 4,898 samples. The seven encountered quality classes and the estimated priors, covariance ranks, and regularization values are summarized below.

Quality	N_k	Prior	Cov. rank	λ_k
3	20	0.004083	11	1.421885
4	163	0.033279	11	0.290785
5	1457	0.297468	11	0.209196
6	2198	0.448755	11	0.180003
7	880	0.179665	11	0.115158
8	175	0.035729	11	0.124062
9	5	0.001021	4	0.117890

2.1 Classification Results

$$\hat{P}_e = \frac{2939}{4898} = 0.60004083 = 60.0041\%$$

The classifier made 2,939 errors out of 4,898 samples, corresponding to 60.0041% estimated error and 39.9959% training accuracy.

Confusion matrix counts (rows = true quality, columns = predicted quality):

True \ Pred.	3	4	5	6	7	8	9
3	5	2	5	5	3	0	0
4	2	0	65	66	30	0	0
5	1	1	322	803	328	2	0
6	1	0	170	927	1095	5	0
7	0	0	23	152	698	7	0
8	0	0	0	30	139	6	0
9	0	0	0	0	4	0	1

Most errors occur among adjacent quality scores. In particular, 803 quality-5 samples were classified as 6, 1,095 quality-6 samples were classified as 7, and 139 quality-8 samples were classified as 7. This indicates strong overlap between neighboring wine-quality classes.

2.2 PCA Visualization and Gaussian-Model Suitability

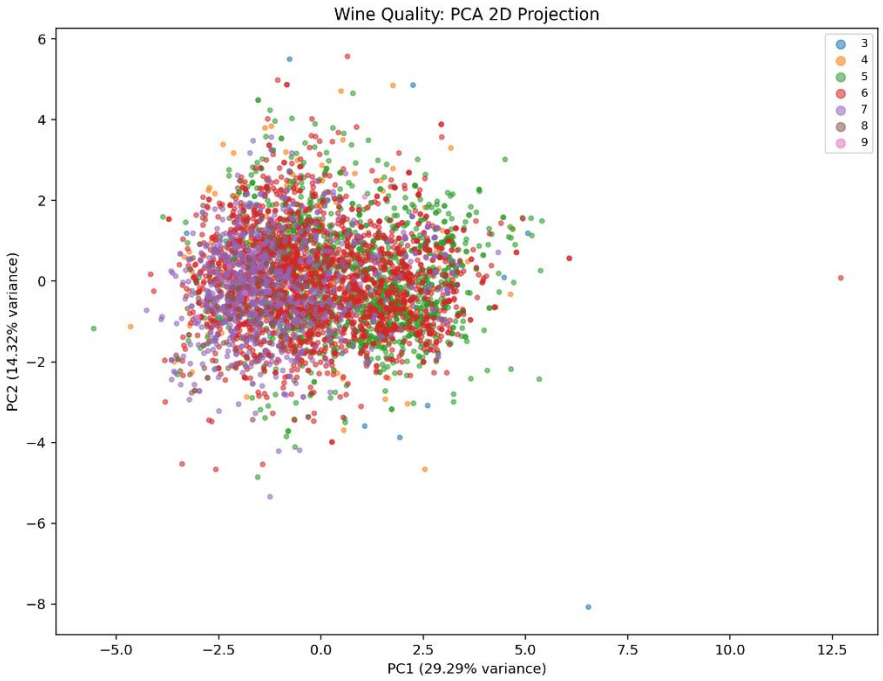


Figure 1. Wine Quality 2D PCA projection. PC1 explains 29.29% and PC2 explains 14.32% of the standardized-data variance.

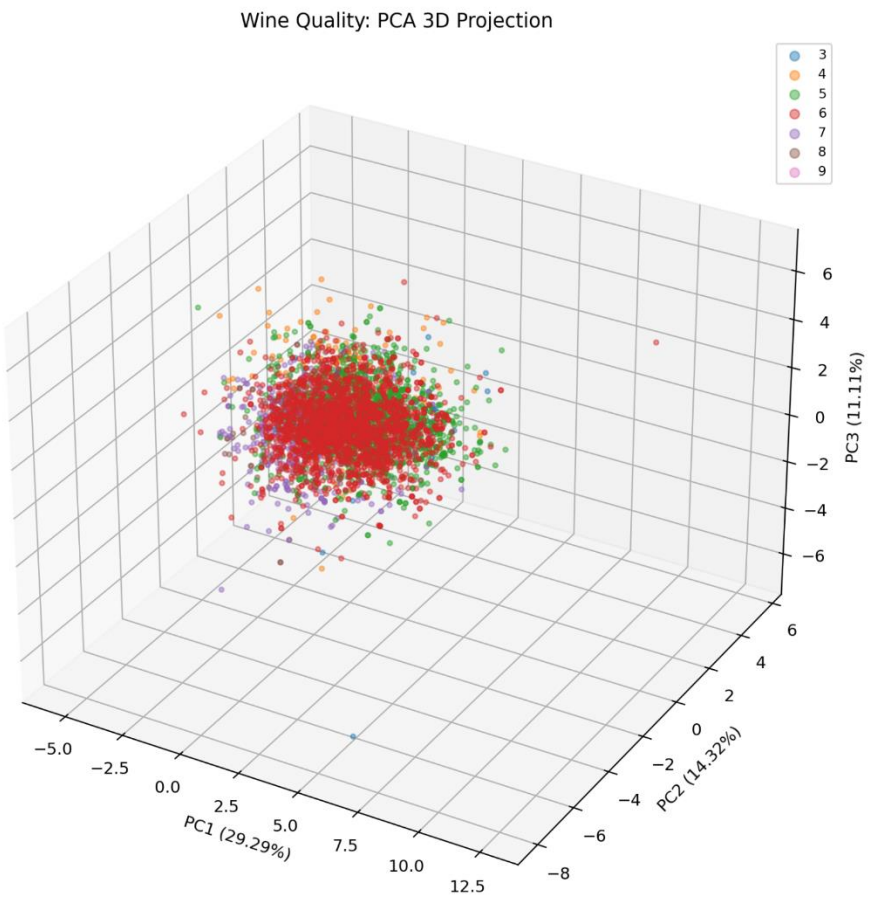


Figure 2. Wine Quality 3D PCA projection. PC3 explains an additional 11.11% of the standardized-data variance.

The first three principal components explain approximately 54.72% of the standardized-data variance. Both the 2D and 3D projections show extensive overlap among the wine-quality classes, especially qualities 5, 6, and 7. The classes do not appear as clearly separated, compact Gaussian clusters.

Therefore, a single Gaussian per quality level is only a rough approximation for this dataset. The strong class overlap and possible non-Gaussian or multimodal structure help explain the relatively large 60.00% training error. The severe class imbalance also affects parameter reliability; for example, quality 9 contains only five samples.

3. Human Activity Recognition Dataset

The HAR dataset contains 10,299 samples, 561 features, and six activity classes. All class covariance estimates are rank deficient relative to the 561-dimensional feature space, which justifies regularization. In the table, W = WALKING, WU = WALKING_UPSTAIRS, WD = WALKING_DOWNSTAIRS, SIT = SITTING, STD = STANDING, and LAY = LAYING.

Activity	N_k	Prior	Cov. Rank	λ_k
W	1722	0.167201	469	4.285584e-05
WU	1544	0.149917	468	4.402428e-05
WD	1406	0.136518	469	6.041423e-05
SIT	1777	0.172541	467	3.418392e-05
STD	1906	0.185067	467	3.084661e-05
LAY	1944	0.188756	466	4.290347e-05

3.1 Classification Results

$$\hat{P}_e = \frac{85}{10299} = 0.00825323 = 0.8253\%$$

The classifier made 85 errors out of 10,299 samples, giving an estimated error probability of 0.8253% and training accuracy of 99.1747%.

Confusion matrix counts (W = WALKING, WU = WALKING_UPSTAIRS, WD = WALKING_DOWNSTAIRS, SIT = SITTING, STD = STANDING, LAY = LAYING):

True \ Pred.	W	WU	WD	SIT	STD	LAY
W	1722	0	0	0	0	0
WU	0	1544	0	0	0	0
WD	0	12	1394	0	0	0
SIT	0	0	0	1707	70	0
STD	0	0	0	3	1903	0
LAY	0	0	0	0	0	1944

All 85 errors are concentrated among physically similar activities: 12 WALKING_DOWNSTAIRS samples were classified as WALKING_UPSTAIRS, 70 SITTING samples were classified as STANDING, and 3 STANDING samples were classified as SITTING.

3.2 PCA Visualization and Gaussian-Model Suitability

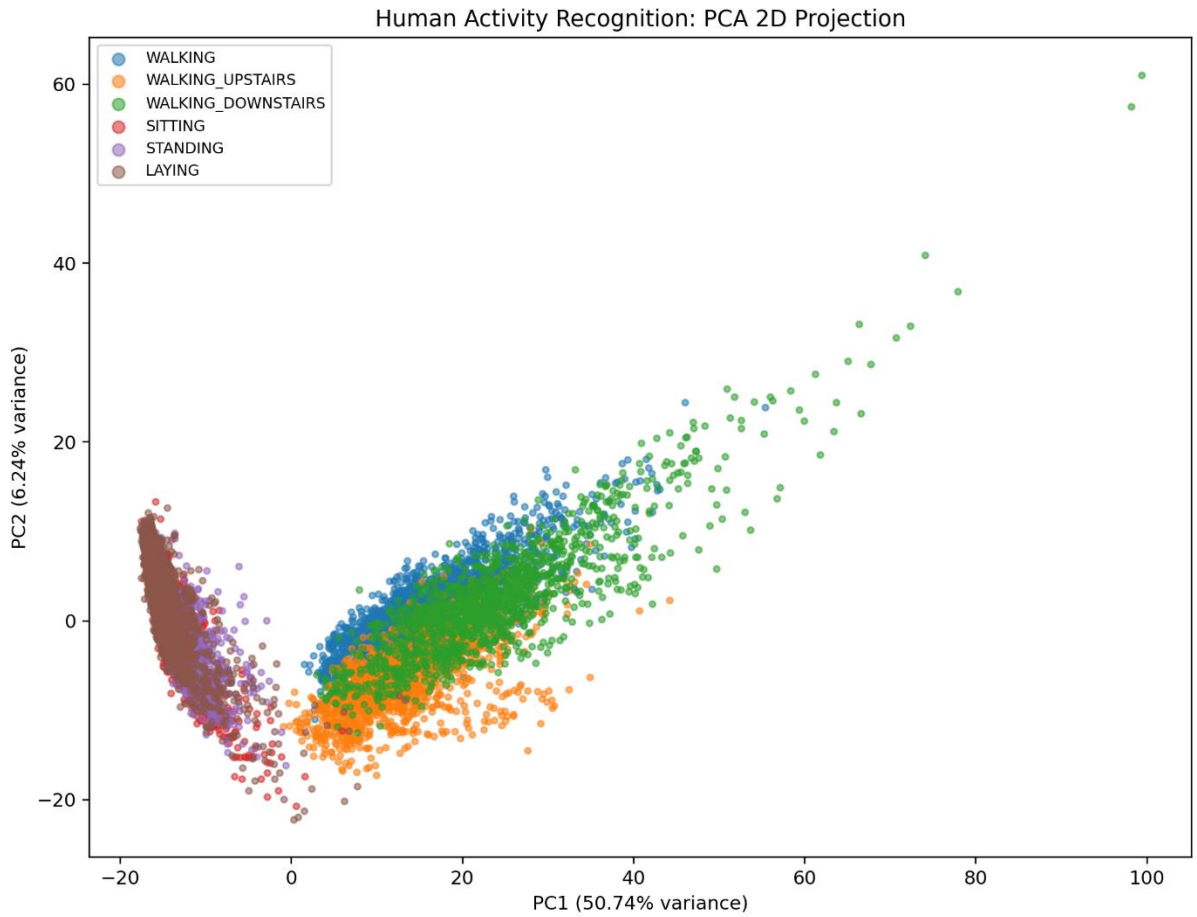


Figure 3. HAR 2D PCA projection. PC1 explains 50.74% and PC2 explains 6.24% of the standardized-data variance.

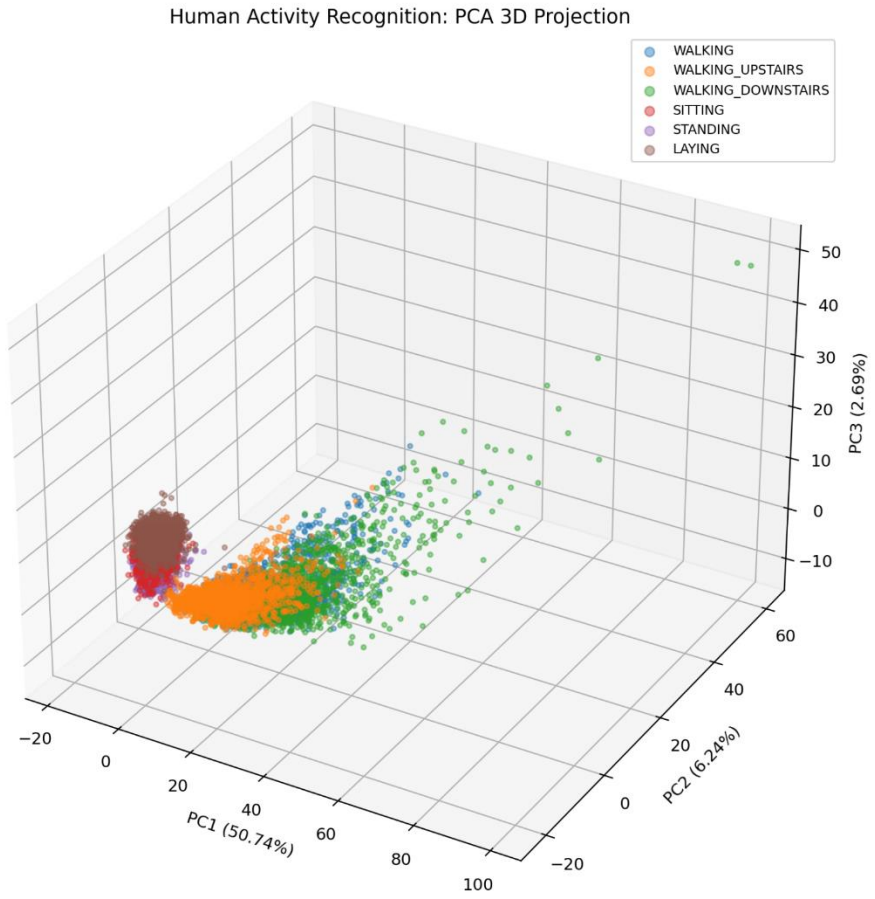


Figure 4. HAR 3D PCA projection. PC3 explains an additional 2.69% of the standardized-data variance.

The first three principal components explain approximately 59.67% of the standardized-data variance. The PCA projections show a broad separation between many dynamic and static activities, but substantial overlap remains in the first few principal components. In particular, SITTING and STANDING occupy very similar regions, which is consistent with the main confusion-matrix error.

The very low training error shows that the class-specific full-covariance Gaussian model is highly effective for the HAR feature vectors when all 561 dimensions are used. The overlap visible in 2D and 3D PCA does not contradict the high classification accuracy because PCA is an unsupervised low-dimensional projection and discards more than 40% of the variance. The full classifier uses all 561 features.

The result should still be interpreted as a training error. Because a separate high-dimensional covariance matrix is estimated for every activity and evaluated on the same samples used for training, the 0.8253% error may be optimistic relative to performance on unseen data.

4. Summary and Comparison

Dataset	Samples	Features	Errors	P(error)	Training accuracy
Wine Quality	4898	11	2939	60.0041%	39.9959%
HAR	10299	561	85	0.8253%	99.1747%

The same minimum-probability-of-error Gaussian framework behaves very differently on the two datasets. For Wine Quality, extensive class overlap makes a single Gaussian per class a weak approximation and produces a high error probability. For HAR, the full high-dimensional feature representation allows the Gaussian classifier to separate the training samples extremely well, with the remaining errors concentrated among closely related activities.

Overall, the required Gaussian models, parameter estimates, covariance regularization, minimum-P(error) classification rule, confusion matrices, empirical error probabilities, and 2D/3D PCA visualizations have all been implemented and reported.